Peoplesoft ID:

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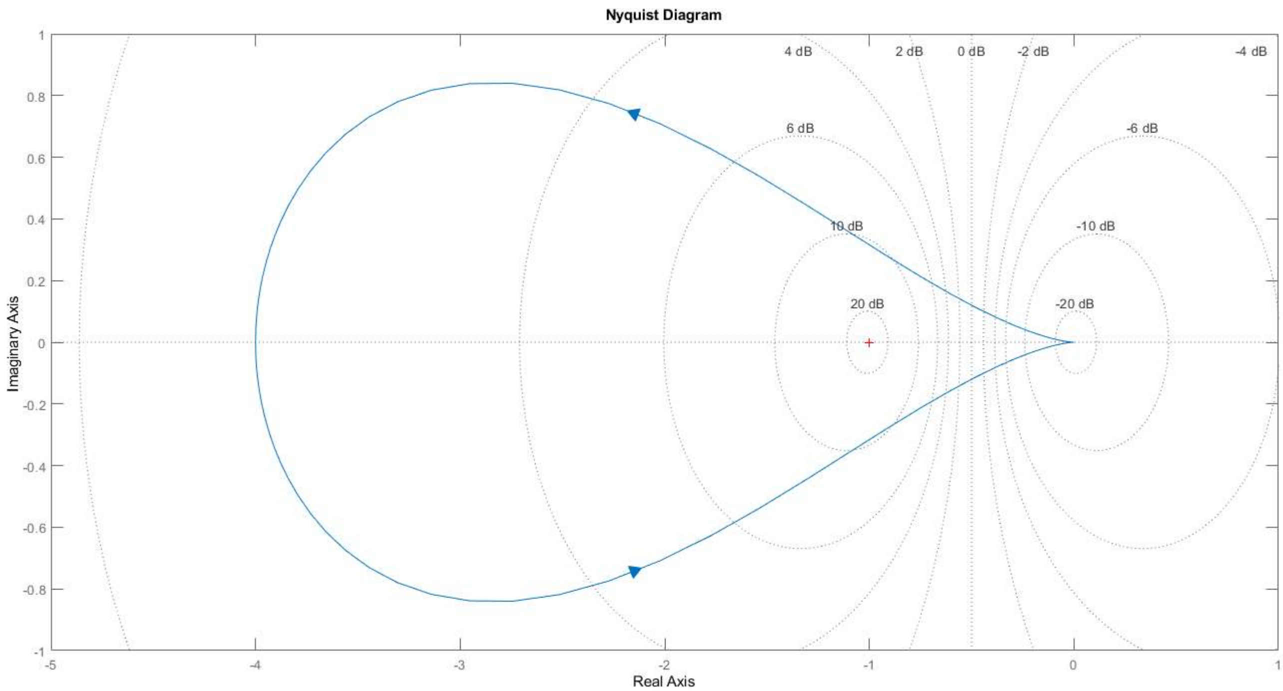
EEE3094S 2023: Deferred Exam

- The exam is open book
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write "see attached pages" by the question) and make sure they are marked with your peoplesoft ID.
- There are five questions totalling 100 marks.
- You have 15 minutes reading time and 3 hours to complete the exam.
- The last page is an information sheet.

Question 1 [15]

The Nyquist plot below shows the frequency response of a second-order system that has one unstable open-loop pole. Based on this response, decide whether each of the following statements applies to the system. Decide whether each of the following statements about the system is TRUE or FALSE. Briefly explain your decisions.

You may wish to draw on or annotate the plot to assist your working or draw attention to key features. A larger version of the plot is provided after the answer spaces.
[5 x 3 marks]



a) The system is stable in closed loop.

b) The closed-loop response is oscillatory.

c) The system is type 1.

d) The closed-loop system will amplify noise.

e) The open-loop system will amplify noise.

a)

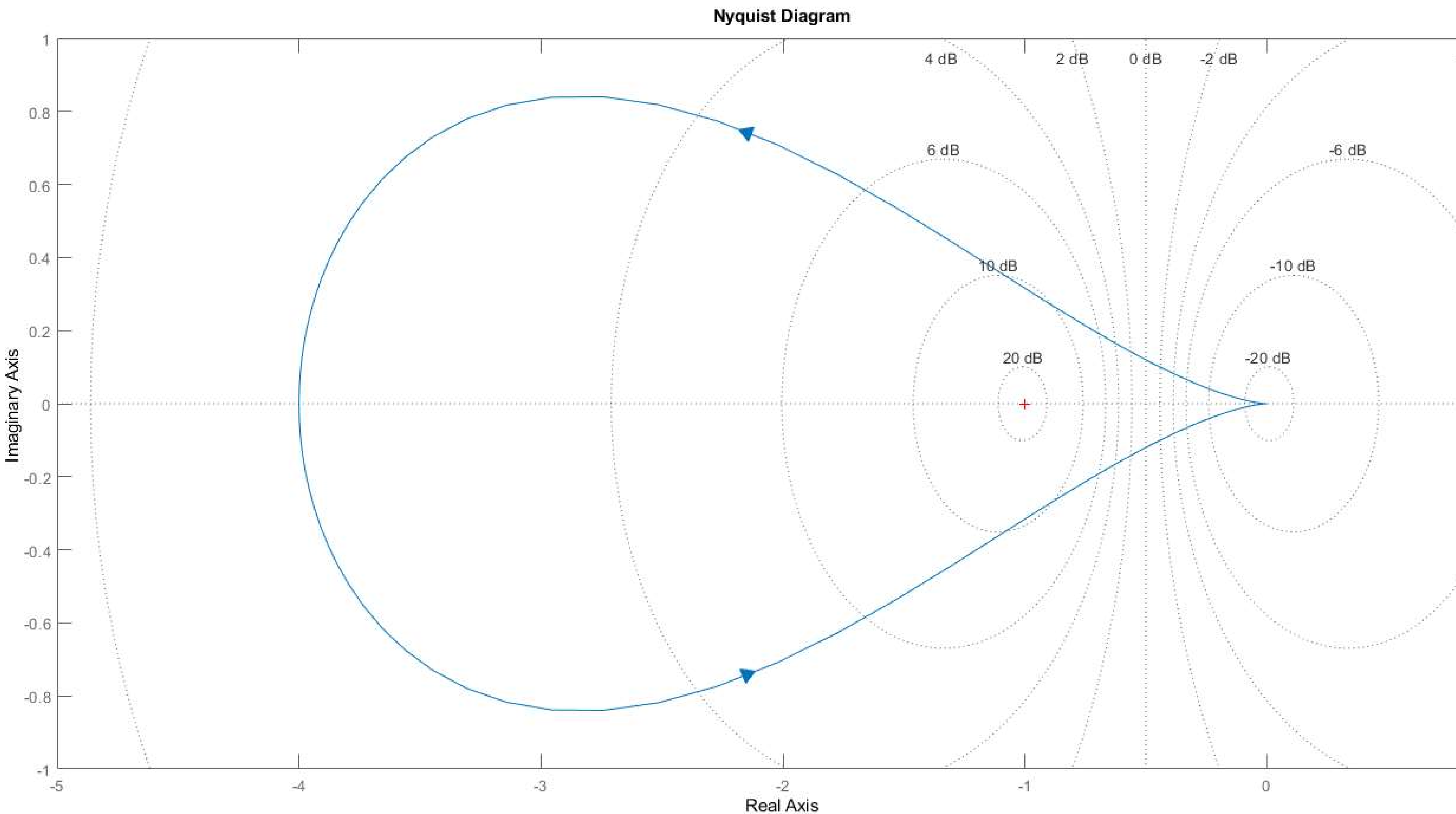
b)

c)

d)

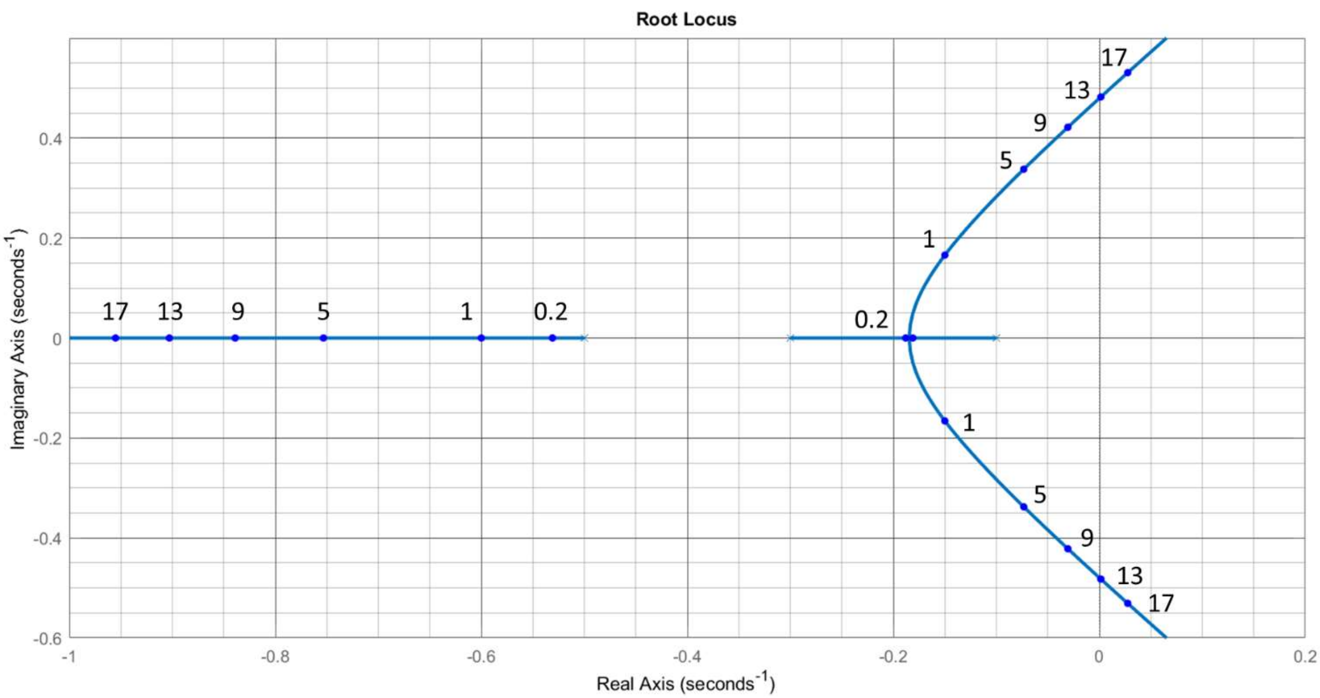
e)

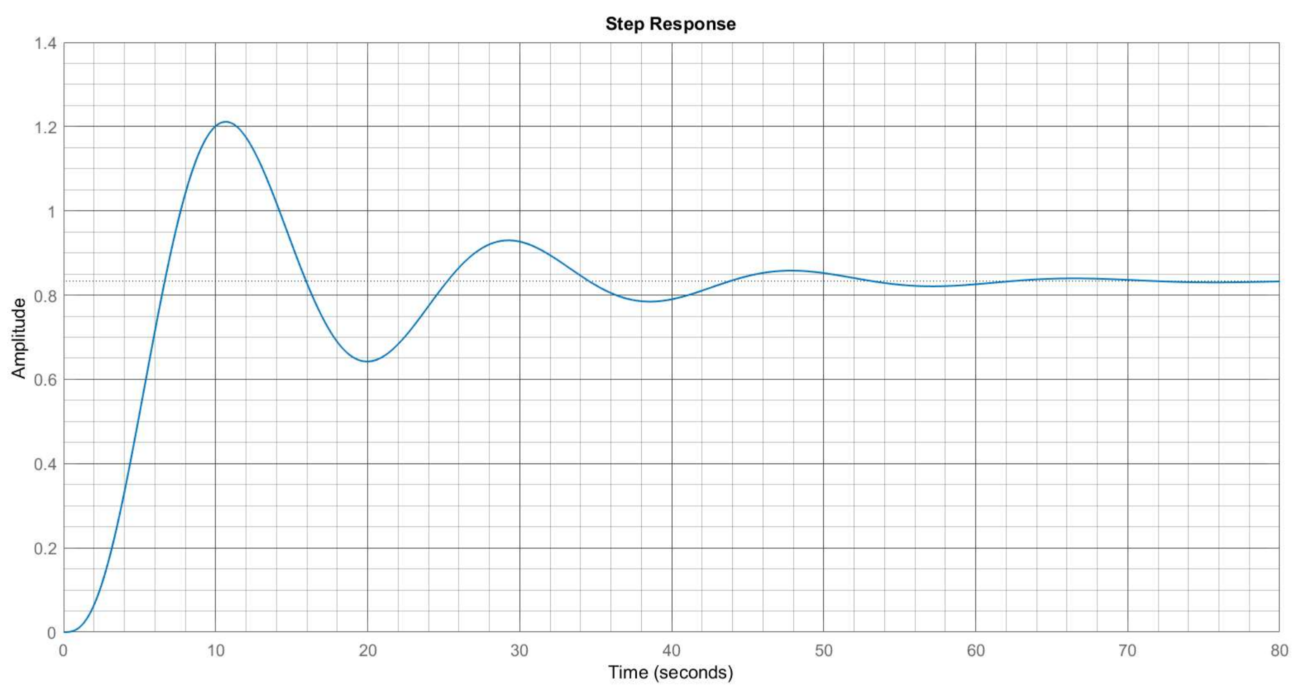
Extra space


Nyquist Diagram
4 dB
2 dB
0 dB
-2 dB
6 dB
-6 dB
10 dB
-10 dB
20 dB
-20 dB
Imaginary Axis
Real Axis
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-5
-4
-3
-2
-1
0

Question 2 [15]

The plots below show the annotated root locus and closed-loop step response of a third-order system in negative feedback with a proportional controller. The DC gain of the uncontrolled system is 1. The numbers indicate the proportional gain of the closed-loop system corresponding to the poles on the root locus.





a) Use these plots to identify the position of the closed-loop poles and hence, the gain of the controller. [5]

b) What is the gain margin of the system? [2]

c) Suggest a reduced-order model for the closed-loop system. [2]

d) Would you recommend modeling the closed-loop system as second-order at all gain values? If not, use the annotations on the root locus to suggest a rough range of values where such a model would not be applicable, and explain why. [3]

e) Briefly describe how the settling time, damping and frequency of the closed-loop response will be affected as the gain is increased. [3]

a)

b)

c)

d)

e)

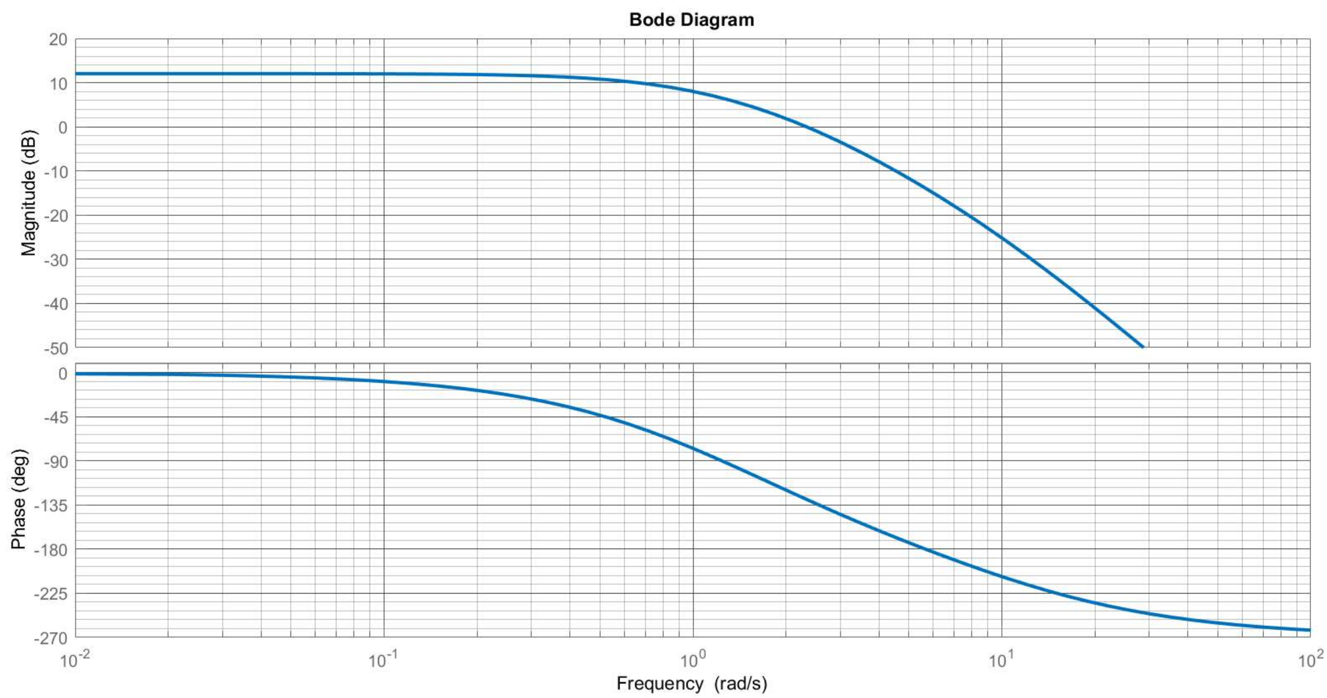
Extra space:

Question 3 [10]

Consider the system $G(s) = \frac{80}{(s+1)(s+2)(s+10)}$. You wish to place this system in feedback with a controller to obtain the following performance specifications:

- Settling time
- Damping ratio greater than 0.5
- Tracking accuracy of greater than 97.5% for position setpoints.

The Bode plot of the system is shown below. A larger version of the plot is provided after the answer spaces.



a) Show that the dominant closed-loop pole pair $-1.25 \pm j2.04$ meets the settling time and damping specifications. Can this pole pair be obtained through proportional control? If so, state the required gain. [4]

b) Would you recommend using proportional control to satisfy the accuracy specification? [2]

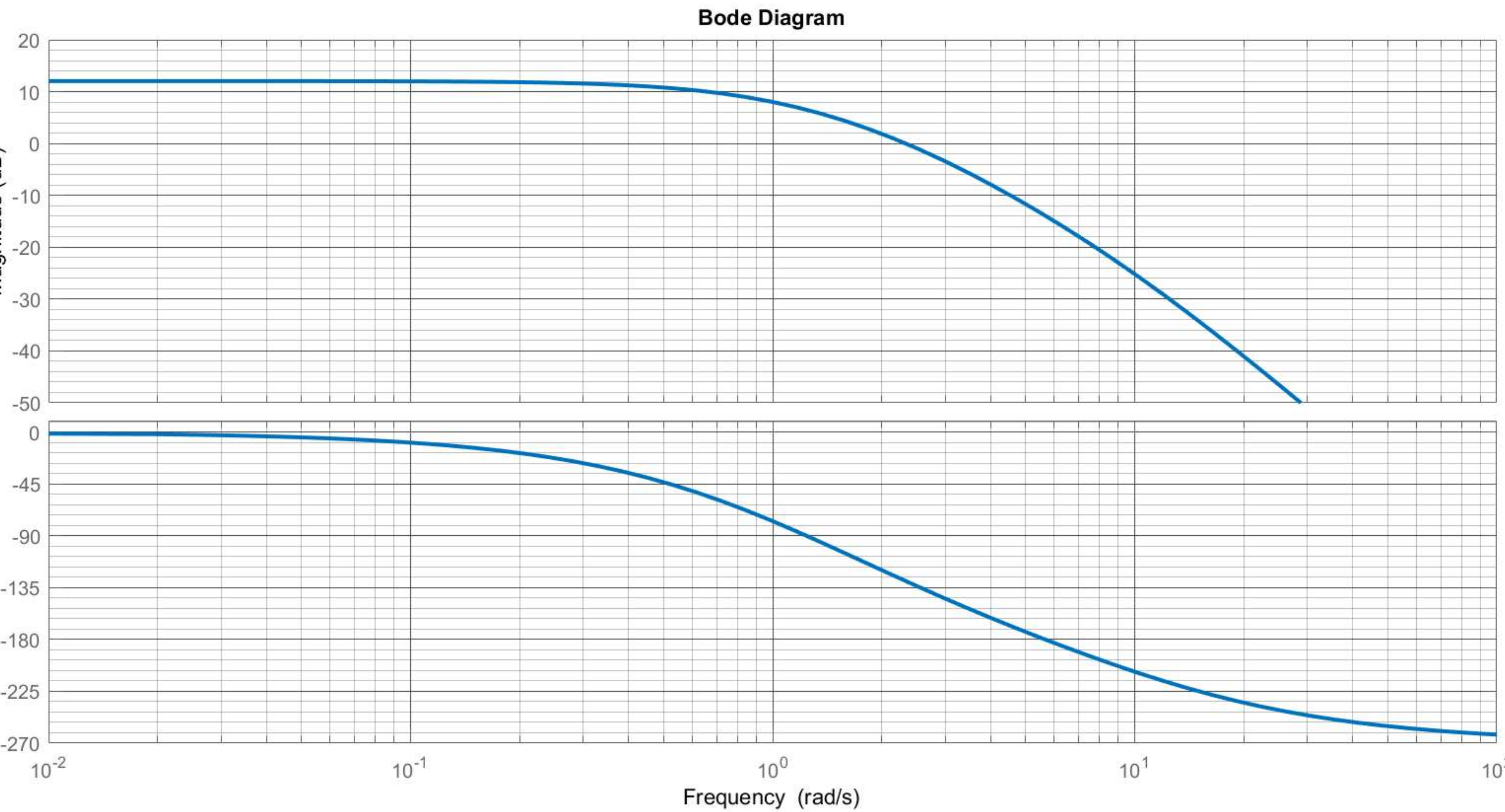
c) Design a lag compensator to obtain the desired pole pair in (a) while satisfying the accuracy specification [4].

a)

b)

c)

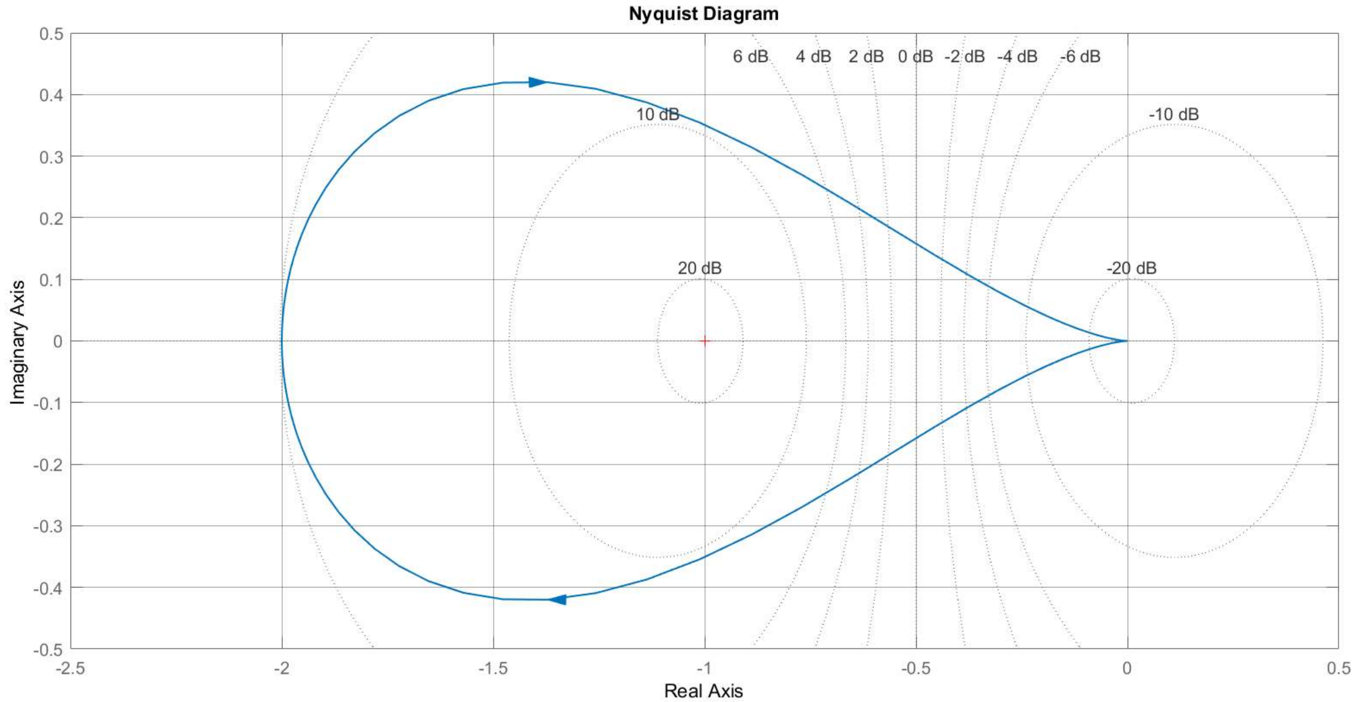
Extra space:



Bode Diagram
20
10
0
-10
-20
-30
-40
-50
0
-45
-90
-135
-180
-225
-270
10^-2
10^-1
10^0
10^1
10^2
Frequency (rad/s)

Question 4 [30]

The following Nyquist plot shows the frequency response of a system with one unstable open-loop pole. The Bode plot and Nichols chart are provided on the next page.



a) Is it possible to stabilize the system using a proportional controller? If so, determine the required gain. [2]

b) You decide to stabilize the system using a lead compensator instead. The final design should satisfy the following specifications:
   i. Tracking accuracy of 90% for a constant setpoint.
   ii. Damping ratio exceeding 0.7
   iii. Settling time (to within 5% of the final value) in less than 6 seconds.

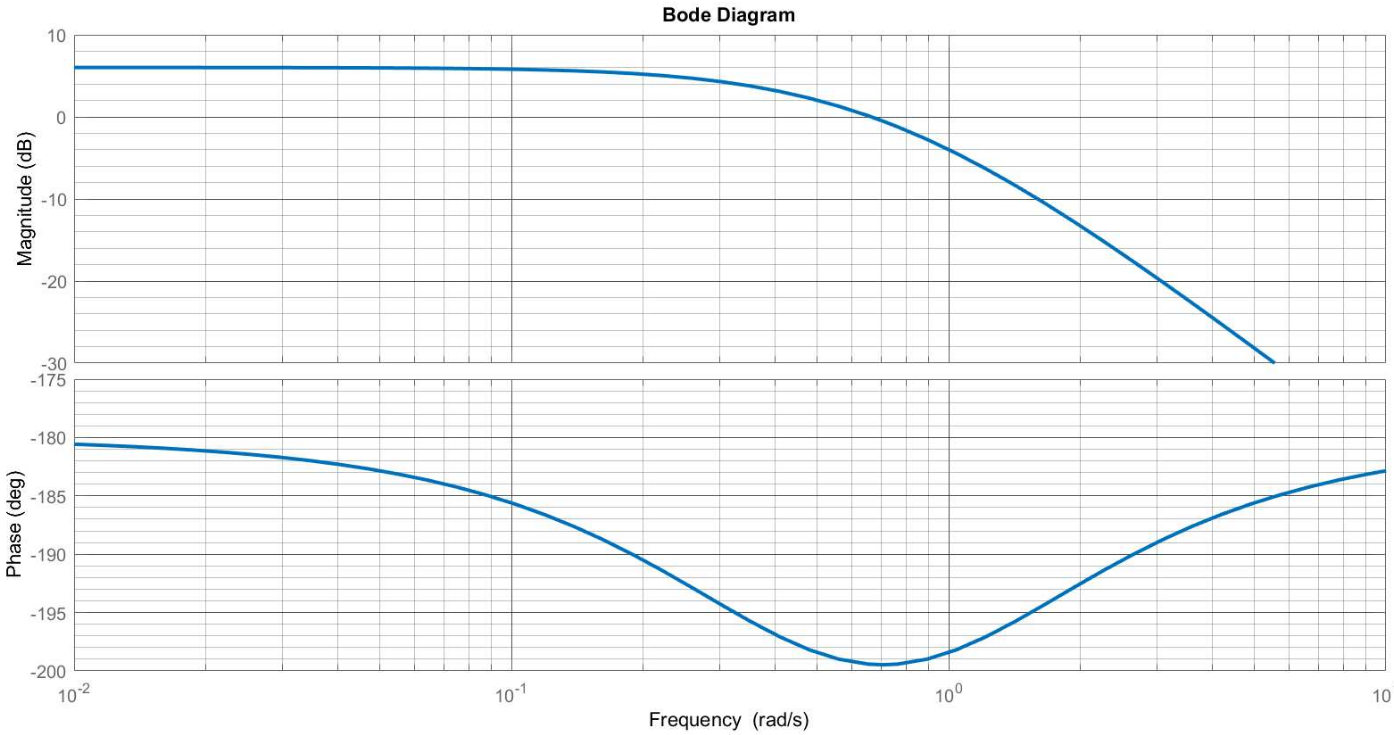
   Interpret each of these specifications in frequency domain terms, and draw them on the Nichols chart. [10]

c) Show that an increase in gain of 14 dB is sufficient to satisfy the accuracy specification. Sketch the response on the Nichols chart of the system with this additional gain. [2]

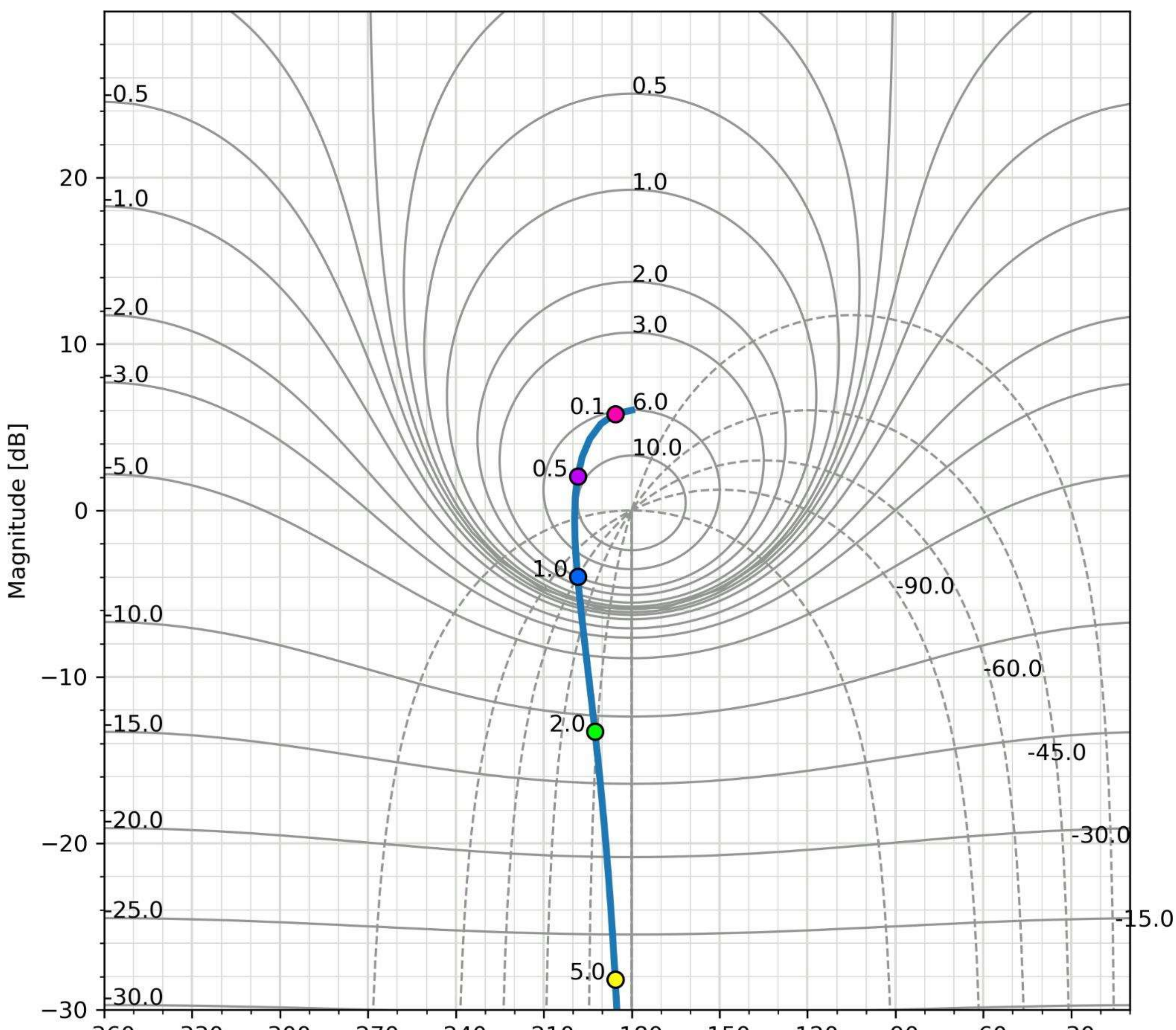
d) Based on the sketch in (c), is it possible to satisfy the damping specification with a physically realizable compensator at this gain? If not, suggest how the design could be modified to do so. [4]

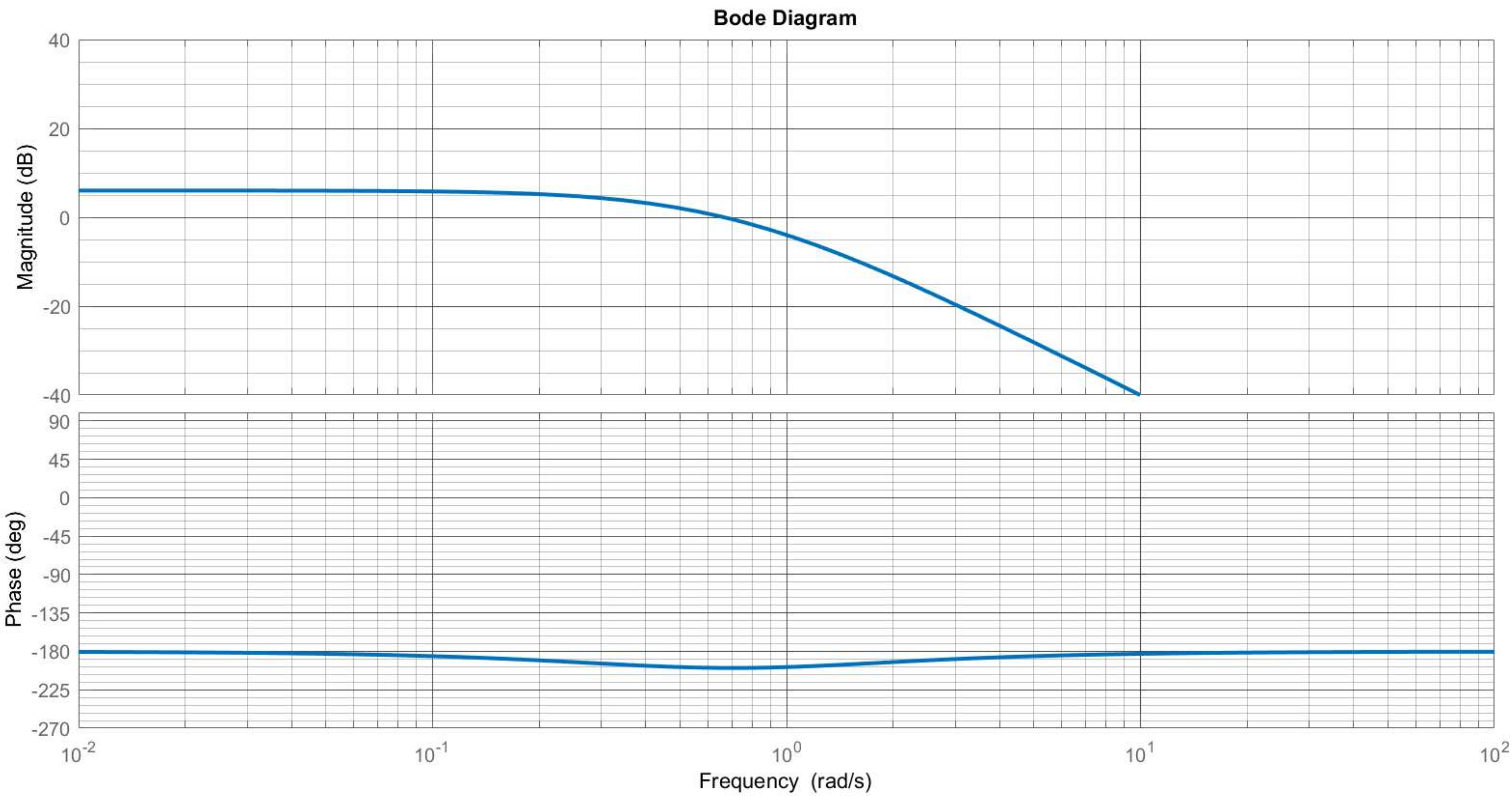
e) You decide to implement a compensator with a pole that has a corner frequency 10 times higher than that of the zero. Determine a suitable midpoint frequency and hence, the transfer function of the complete compensator. [4]

f) Sketch the frequency response of this compensator on the Bode plot, and hence, roughly sketch the response of the compensated system. Use this sketch to evaluate the performance of the compensator with respect to the given specifications, and stability of the system. You may wish to transfer the compensated response to the Nichols chart. (Hint: use the design frequencies noted on the Nichols chart as a reference.) [8]



A larger version of the bode plot with space for sketching the compensator response is provided on the next page.




Bode Diagram
Magnitude (dB)
40
20
0
-20
-40
Phase (deg)
90
45
0
-45
-90
-135
-180
-225
-270
10^-2
10^-1
10^0
10^1
10^2
Frequency (rad/s)

a)

b)

c)

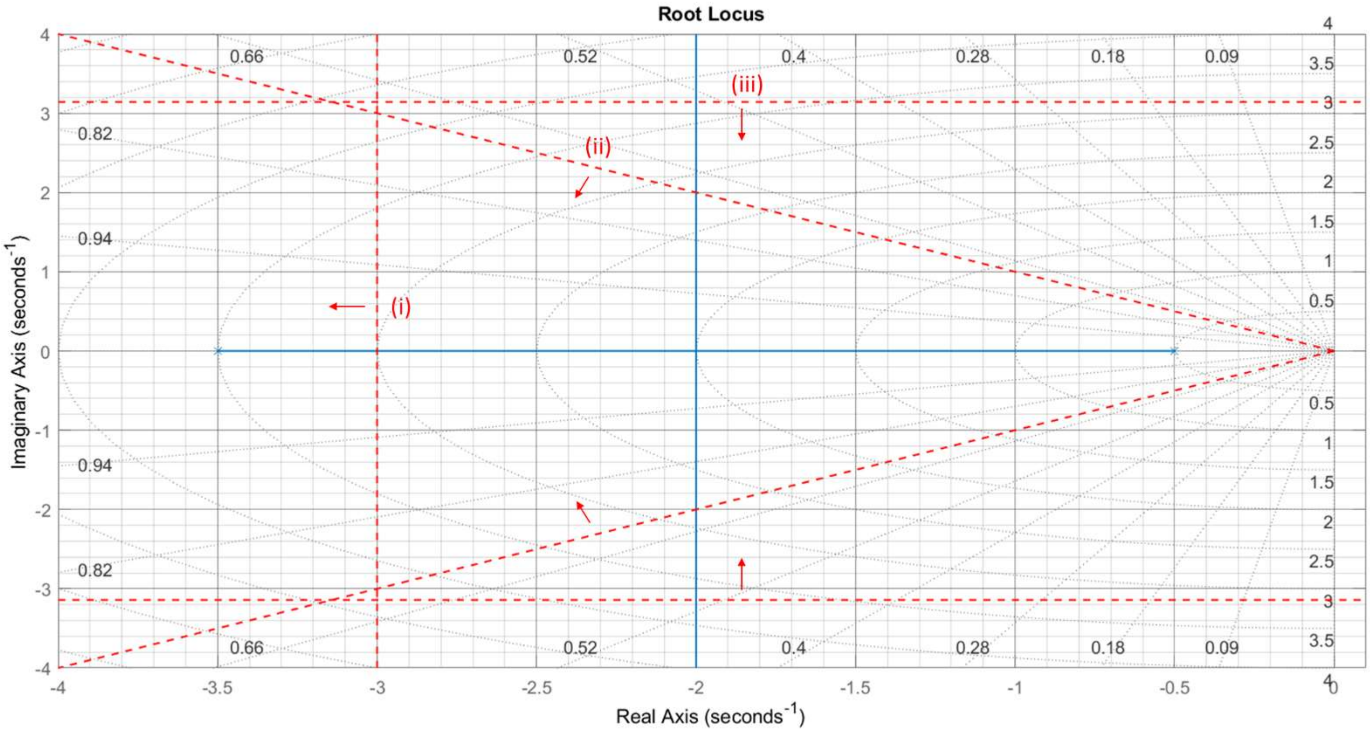
d)

e)

f)

Question 5 [30]

Consider the system $G(s) = \frac{1}{(s+0.5)(s+3.5)}$, with root locus shown below:



a) The dashed lines represent design specifications. What do each of these mean in terms of the required behaviour of the system? Answer in terms of the settling time (to 5%), the amount of overshoot and the frequency of the response. [6]

b) It is important that the final design does not have a resonant frequency between 3 and 3.5 rad/s. Sketch this specification on the root locus. [2]

c) For each specification (a.i – a.iii, b), comment on whether it can be satisfied using proportional control. [4]

d) Consider the following complex pole pairs:
1) $-3 \pm j1.8$
2) $-3.2 \pm j2.7$
3) $-4 \pm j0.1$
Which would you consider to be the best choice for the closed-loop poles of this system? Motivate your answer by describing the disadvantages of the other options. [3]

e) You decide to try placing the dominant closed-loop poles at $-3.1 \pm j2.8$ using a lead compensator. Suggest suitable maximum and minimum values for the placement of the compensator zero. [4]

f) Consider the following options for the position of the zero:
1) -3.5
2) -5.5
3) -7
Calculate the corresponding location of the compensator pole for each one. Which option would you recommend? Motivate your answer by describing the disadvantages of the other options. [8]

g) If the zero at -5.5 is selected, what is the required gain to obtain the desired closed-loop pole pair? What is the steady-state error with respect to a constant setpoint at this value (as a percentage)? [3]

a)

b) (on plot)

c)

d)

e)

f)

f cont.)

g)

Extra space

Extra Space

Information

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$\cos\theta = \zeta$$

$$t_p = \frac{\pi}{\omega_d}$$

$$y_p = A\left(1 + e^{-\zeta\pi/\sqrt{1-\zeta^2}}\right)$$

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_{k\%} = \frac{\ln\left(\frac{k}{100}\sqrt{1-\zeta^2}\right)}{-\zeta\omega_n}$$

$$\%OS = 100e^{\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}\right)}$$

Laplace transforms of signals:

$\boldsymbol{y(t)}$	$\boldsymbol{Y(s)}$	Comments
$\sigma(t)$	$\frac{1}{s}$	unit step
$t\sigma(t)$	$\frac{1}{s^2}$	unit ramp
$e^{-at}\sigma(t)$	$\frac{1}{(s+a)}$	decaying exponential if $a > 0$
$t^n e^{-at}\sigma(t)$	$\frac{n!}{(s+a)^{(n+1)}}$	n^{th} order exponential
$\sin(\omega t)\sigma(t)$	$\frac{\omega}{s^2+\omega^2}$	sinusoid
$\cos(\omega t)\sigma(t)$	$\frac{s}{s^2+\omega^2}$	cosinusoid
$e^{-at}\sin(\omega t)\sigma(t)$	$\frac{\omega}{(s+a)^2+\omega^2}$	damped sinusoid
$e^{-at}\cos(\omega t)\sigma(t)$	$\frac{s+a}{(s+a)^2+\omega^2}$	damped cosinusoid